

Reproductive biology of blue marlin *Makaira nigricans* around Yonaguni Island, southwestern Japan

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Received: 12 May 2008 / Accepted: 29 July 2008 / Published online: 11 December 2008
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Abstract We studied the spawning seasonality and gonadal development of blue marlin *Makaira nigricans* using specimens captured around Yonaguni Island in southwestern Japan between February 2003 and February 2006. The mean ($\pm$ SD) lower jaw–fork length of females (234 ± 24 cm) was greater than that of males (191 ± 12 cm). The smallest mature female and male were 183 and 160 cm, respectively. Most of the 717 females had immature ovaries. However, in March and from May to September, the ovaries of 26 females contained oocytes with yolk globules, hydrated oocytes, or postovulatory follicles. Most males had testes with a large amount of spermatozoa throughout the year. The occurrence of mature blue marlin at Yonaguni Island suggests that spawning occurs here. The mean condition factors (fatness of the fish) of both sexes decreased from March to June or July, presumably as they expended energy to reproduce. We discuss our results in the context of migration theory for blue marlin in the western North Pacific Ocean.

Keywords Blue marlin · Condition factor · *Makaira nigricans* · Migration · North Pacific Ocean · Spawning season

Introduction

The blue marlin *Makaira nigricans* is a large istiophorid billfish that can weigh over 906 kg [1]. The known distribution of this species includes the tropical to temperate waters of the Indo-Pacific and Atlantic Oceans. The Indo-Pacific population was thought to be a distinct species, *M. mazara*, which was isolated from the Atlantic population [1]. However, *M. mazara* is now considered a junior synonym of *M. nigricans* based on molecular analyses [2–4]. Although blue marlin migrate through the Pacific Ocean, there is no evidence of stock structure in the area [5, 6]. The migration patterns of blue marlin have been estimated from seasonal catches by longline fisheries [7, 8]. However, the actual migration routes of individual fish throughout their life histories have not been determined.

The blue marlin is an important commercial and recreational fish species [1]. In Japan, blue marlin was initially fished by local harpoon operations [9]. The species is now primarily fished using a coastal bait-trolling technique, especially around Okinawa Prefecture. Blue marlin are also caught as bycatch by pelagic longline fisheries that target tuna *Thunnus* spp. and swordfish *Xiphias gladius*. The stock of blue marlin in the Pacific Ocean appears to have stayed at approximately the same level since the 1970s [10]. However, further knowledge of biological life history parameters such as reproductive biology is now required for stock assessment analyses. Blue marlin migrate latitudinally [7], a behavior thought to be related to spawning and feeding activity. Accurate determination of their

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maturing season and spawning grounds is key to understanding their migration patterns.

Previous studies on the reproductive biology of blue marlin in the eastern Pacific Ocean [11], western Indian Ocean [12, 13], Atlantic Ocean [14, 15], and Hawaiian waters [16] focused on determining spawning season and size at sexual maturity. However, in the western North Pacific, where most blue marlin populations are concentrated, these biological parameters are poorly understood [17, 18]. Between March and September, blue marlin occur in high abundance around Yonaguni Island, and some blue marlin are found in this region in other months [9]. Active feeding by blue marlin has been observed around Yonaguni Island [19], but no evidence of reproductive behavior was detected in this area. The occurrence of blue marlin larvae indicates the presence of a major spawning ground in the western North Pacific east of the Philippines, and a few larvae have also been collected near Taiwan [20, 21]. Seasonal sex ratios, size at maturity, batch fecundity, and spawning frequency are also important parameters that are needed for accurate stock assessment. We investigated the reproductive seasonality and gonadal development of blue marlin near Yonaguni Island in southwestern Japan.

Materials and methods

Sampling methods

Blue marlin catch data and gonad samples were collected at the fishing port of Kubura on Yonaguni Island (24°27'N, 122°57'E) from 20 February 2003 to 14 February 2006 (Fig. 1). On Yonaguni Island, most fishermen operate a coastal bait-trolling fishery for marlin throughout the year. Details about the fishing methods used at Yonaguni Island are described in a previous report on the feeding habits of blue marlin [19]. There were 35 fishing boats (1.5–5.5 t) registered at Yonaguni Island during 2003–2006, all of which landed their catches at Kubura.

The number of fishing boats operating each day was recorded from February 2003 to February 2005. Body weight (BW) and processed weight (PW: body weight without bill, caudal fin, gills, and viscera) of each blue marlin were weighed and recorded to the nearest 1 kg. Lower jaw–fork length (LJFL) and eye–fork length (EFL) of each fish were measured and recorded to the nearest 1 cm. For our analyses, we primarily used LJFL measurements. The bills and caudal fins of some blue marlin were discarded before landing, so these fish were not measured, and their lengths were calculated from their processed weight values. The relationship between LJFL and PW was solved by seeking a least-squares solution for nonlinear equations. The relationships of LJFL–EFL and PW–BW

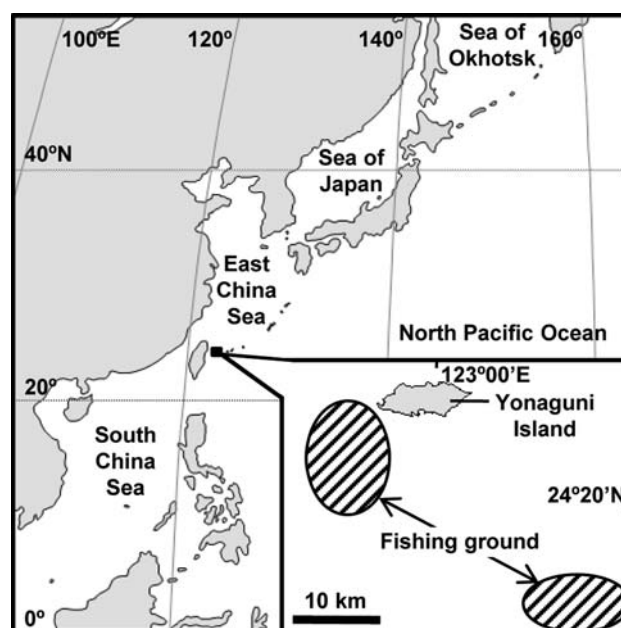


Fig. 1 Location of Yonaguni Island, Japan, where the bait-trolling fishery targets blue marlin *Makaira nigricans* throughout the year

were expressed using linear equations. Ricker's condition factors (CF; $CF = PW \times 10^6 / LJFL^{3.11}$) for 1,262 weighed and measured blue marlin were calculated using the slope (3.11; see "Results") of the length–weight equation [22].

Observation of gonads

Gonads from 1,479 randomly selected blue marlin were removed, and sex was determined by visual inspection of gonad morphology. Gonad weights (GW, kg) of 1,443 blue marlin (959 females and 484 males) were recorded to the nearest 0.01 kg, and the gonadosomatic index (GSI; $GSI = GW \times 100 / PW$) of each fish was calculated.

Samples of tissue from the anterior parts of the gonads of 1,101 fish (717 females and 384 males) landed between February 2003 and February 2005 were collected and fixed in 10% formalin solution. The fixed samples were dehydrated and embedded in paraffin wax. The tissue–paraffin block was cut transversely into 7–10 μ m thick sections, which were mounted on glass slides stained with Mayer's hematoxylin and counterstained with eosin. Oocyte development was staged following previous studies [23–25], and ovarian classifications were based on the most advanced oocyte developmental stage observed in each ovary. An atretic ovary was defined as that in which >70% of the most advanced oocytes were atretic. Females with tertiary yolk and hydrated ovaries were considered reproductively mature. Oocyte diameters were measured at each oocyte stage under the profile projector. The presence or absence of postovulatory follicles (POF) in the ovary was also

recorded. Spawning frequency was determined by the proportion of females with POF to the total number of active females [26]. Testicular classes were determined based on the ratio of spermatozoa area in the sections [27, 28].

Fecundity estimates

Tertiary yolk and hydrated oocyte diameters of mature ovaries in adult fish (232 cm LJFL) caught on 28 May 2004 were measured under the profile projector to define the average diameters of tertiary yolk oocytes and hydrated oocytes. The fecundities of 11 females and the batch fecundities of three females were estimated, where fecundity was defined as the number of tertiary yolk oocytes and hydrated oocytes combined, and batch fecundity was defined as the number of hydrated oocytes only. Three subsamples of approximately 0.1–0.5 g were taken from the ovaries, and the tertiary yolk oocytes and hydrated oocytes of each subsample were teased from their follicular membranes and counted. The mean number of oocytes per gram was used to estimate fecundity and batch fecundity from whole gonad weight. On 4 July 2005, a female with free eggs in the ovarian cavity was landed. The diameters of the free eggs were measured, and the fecundity and the batch fecundity of this female were also estimated.

Results

Monthly landings and body sizes

In total, 2,257 blue marlin were landed at the fishing port of Kubura, and we observed both seasonal and annual variations (Fig. 2). Landings increased in March 2003, January 2004, and April 2005. Landings suddenly decreased in June in all three years, then increased again in July in all three years. They decreased again in October 2003, August 2004, and October 2005. Monthly mean landing number per boat per day (catch-per-unit-effort, CPUE) was high (>0.20) from February to May, July, and September 2003, and from January to May 2004. Monthly mean CPUE was notable in February 2003 and from January to March 2004 (>0.40). The number of females landed during the observation period (986) was double that of males (493). Except when the catches were small, $>50\%$ of the monthly landings consisted of females. Pooled monthly sex ratios did not differ significantly among months (chi-squared test for independence; $df = 11$, $\chi^2 = 9.40$, $P = 0.58$).

The relationships of PW–BW and LJFL–EFL were linear, with high contributing rates ($r^2 = 0.997$, 0.992), and could be used to predict values when the actual measurements were not recorded. LJFL and PW were also highly

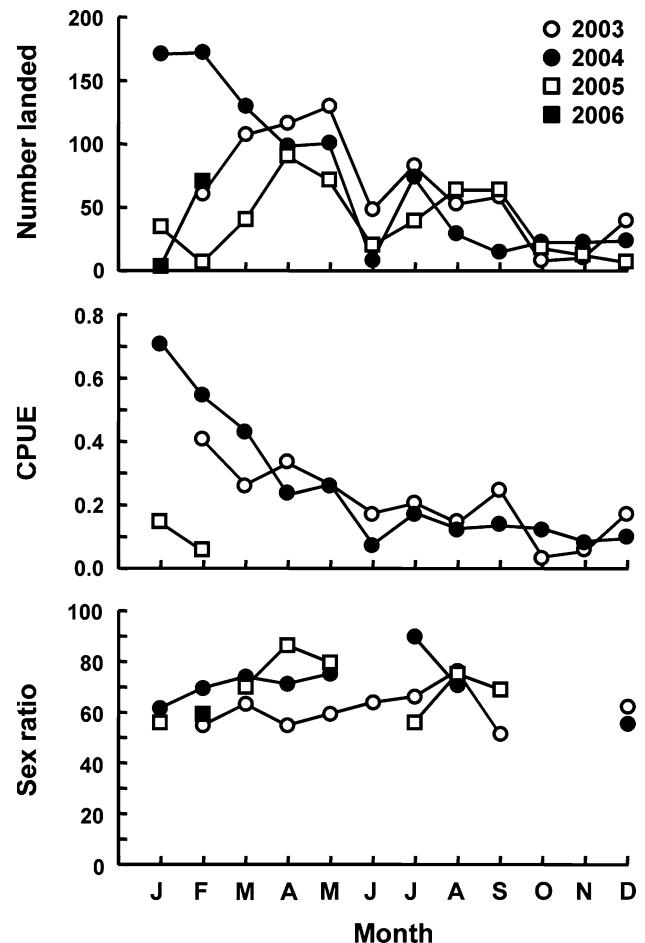


Fig. 2 Monthly changes in the number landed, catch number per boat per day (catch per unit effort, CPUE), and sex ratio (% females) for blue marlin *Makaira nigricans* caught at Yonaguni Island between 20 February 2003 and 14 February 2006. CPUE was not calculated after March 2005. The sex ratios in some months with <10 samples were excluded

correlated, and each could be calculated from the other. PW was proportional to LJFL to the power of 3.11. The equations for these relationships follow:

$$PW = -0.532 + 0.901 \times BW \quad (n = 1,240, r^2 = 0.997) \quad (1)$$

$$LJFL = 7.51 + 1.11 \times EFL \quad (n = 1,304, r^2 = 0.992) \quad (2)$$

$$LJFL = 56.8 \times PW^{0.301} \quad (n = 1,305, r^2 = 0.936) \quad (3)$$

$$PW = 4.70 \times 10^{-6} \times LJFL^{3.11} \quad (n = 1,305, r^2 = 0.936). \quad (4)$$

Females were larger than males. The LJFL of females ranged from 155 to 352 cm (134–320 cm EFL) with a mean ($\pm$ SD) of 234 ± 24 cm (Fig. 3). Most female measurements (98%) were between 180 and 300 cm. Males were 160–233 cm (140–203 cm EFL) with a mean of 191 ± 12 cm. Most measurements (98%) were between

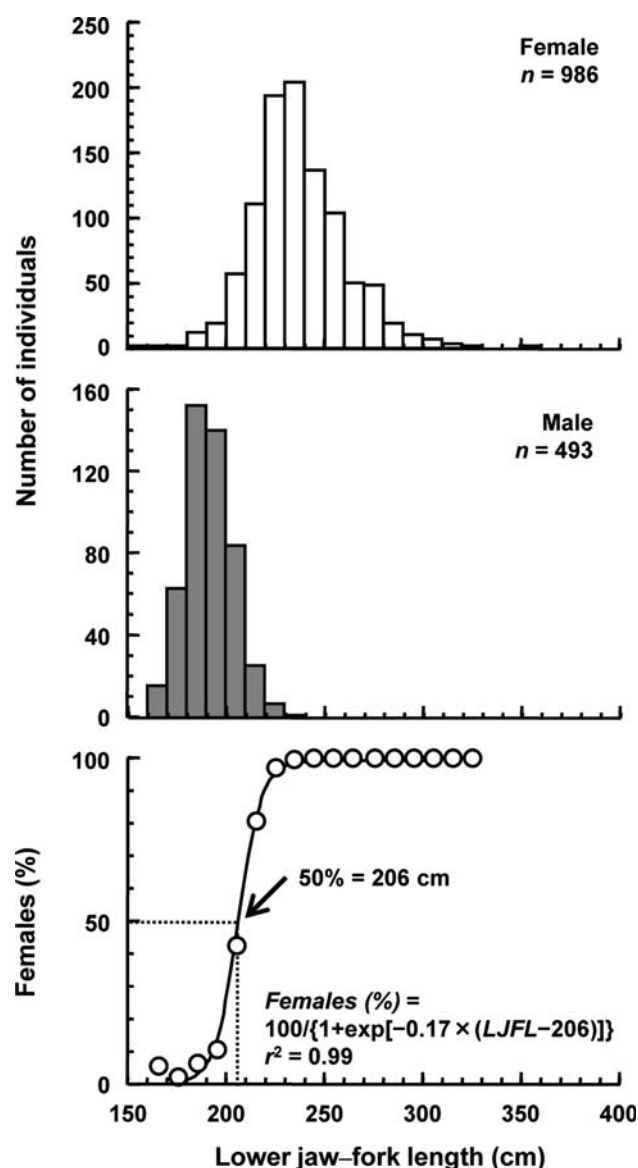


Fig. 3 Length frequency distributions of female and male blue marlin *Makaira nigricans* caught at Yonaguni Island between February 2003 and February 2006. A logistic curve relating the percentage of females to 10-cm lower jaw–fork length (LJFL) classes was fitted using the landing data. Plots showing the percentage of females with <10 samples were excluded from the logistic curve

160 and 220 cm. Females were dramatically larger than males, and the percentage of females increased as LJFL increased. A logistic curve fitted to the percentage of females and LJFL showed that 50% of individuals measuring 206 cm LJFL were female, whereas 100% of individuals above 240 cm LJFL were female.

Classification of gonads

Based on 717 histological ovarian examinations, eight oocyte developmental stages were established. The early

perinucleolus stage (20–60 μm) was the least developed oocyte stage that had a relatively large nucleus (Fig. 4a). The late perinucleolus stage (80–150 μm) was the second undeveloped stage for which cytoplasm stained lighter than the former stage (Fig. 4b). In the yolk vesicle stage (150–200 μm), yolk vesicles appeared midway between the germinal vesicle and periphery in the ooplasm (Fig. 4c). In the primary yolk stage (190–270 μm), yolk began to accumulate, and a small number of yolk granules appeared in the cytoplasm (Fig. 4d). In the secondary yolk stage (230–390 μm), yolk accumulation continued and the yolk globules occupied more than half the area of the cytoplasm (Fig. 4e). In the tertiary yolk stage (380–560 μm), yolk accumulation was almost complete, and yolk globules filled the cytoplasm (Fig. 4f). The hydrated stage (640–880 μm) just before spawning showed the largest oocytes and large, coalesced yolk globules (Fig. 4g). In the atretic ovary, a large number of atretic oocytes of various sizes were present (Fig. 4h). POF were found in the three ovaries in the tertiary yolk stage (Fig. 4i). Combining these eight oocyte stages resulted in six ovarian classes: perinucleolus (early perinucleolus and late perinucleolus), yolk vesicle, early yolk (primary yolk and secondary yolk), tertiary yolk, hydrated, and atretic ovaries.

According to 384 histological testes examinations, four testicular classes based on the percentage of the area occupied by spermatozoa were established: <25% area occupied by spermatozoa and >50% area occupied by spermatogonia (class 1; Fig. 5a); 25–50% spermatozoa and >25% spermatocytes and spermatids (class 2; Fig. 5b); 50–75% spermatozoa and <25% spermatocytes and spermatids (class 3; Fig. 5c); and >75% spermatozoa (class 4; Fig. 5d).

The mean female GSI increased as ovarian maturity approached spawning time (Table 1). Most females containing only unyolked oocytes had a GSI < 0.80, while those with mostly yolked oocytes had a GSI > 1.20, and those with mature ovaries (tertiary yolk and hydrated) had a GSI > 1.80. The mean GSI for females with atretic ovaries was relatively low (1.00), and the mean male GSI was fairly constant across the four testicular maturity classes (0.48–0.53).

Monthly changes in gonad condition

The percent composition of ovarian maturity classes (years pooled) changed seasonally (Fig. 6). Tertiary yolk ovaries were observed in March for one specimen and May–September for 22 specimens. Hydrated ovaries were collected from a single specimen in each of May, June, and September. Mature ovaries comprised only 9.1% (25) of the ovaries collected (274) in May–September. The sex ratio (% females), calculated only from mature specimens, was 16% (25/161) in May–September. POF were found in only

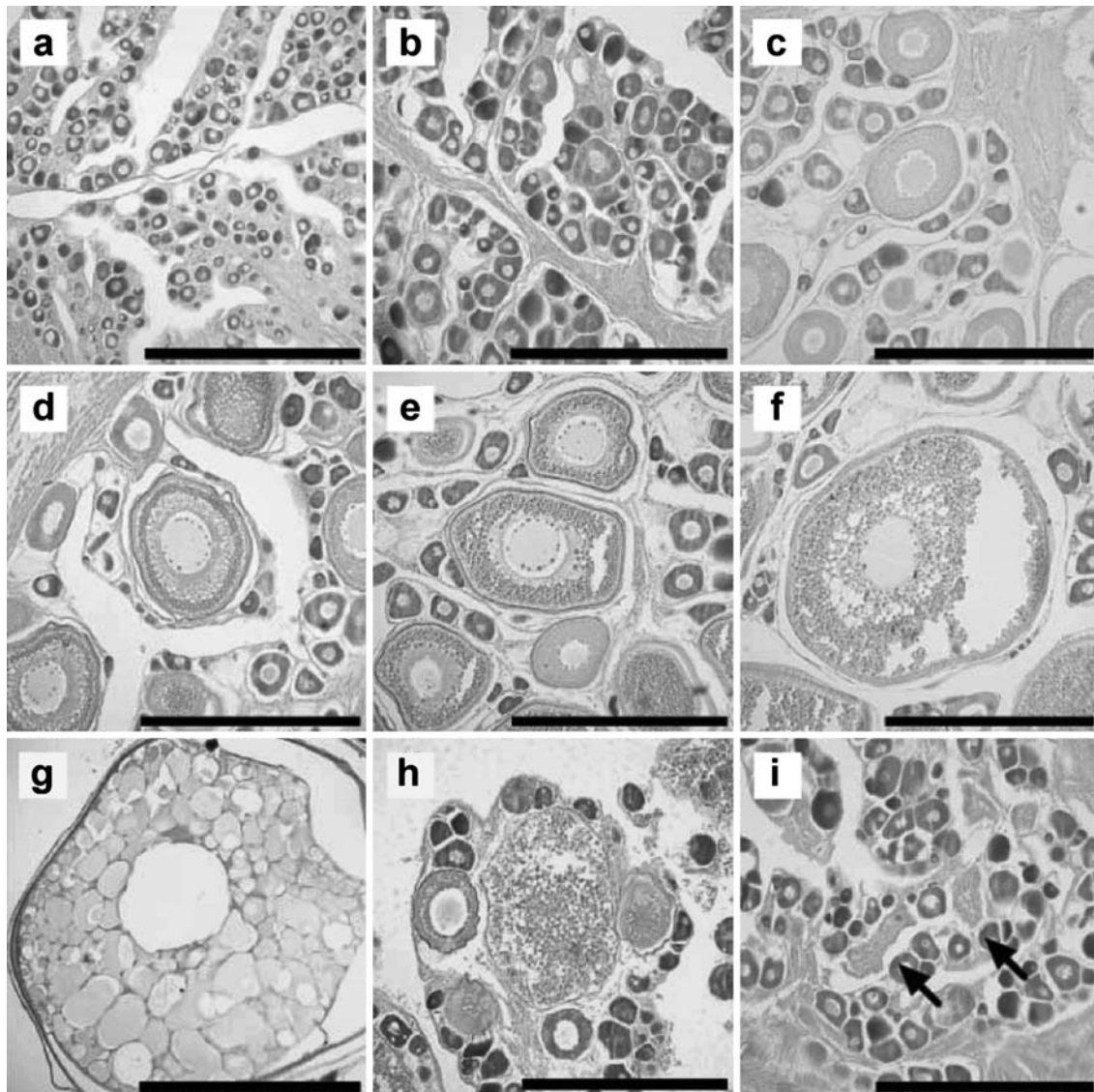


Fig. 4 Photomicrographs of eight oocyte maturity stages and postovulatory follicles for blue marlin *Makaira nigricans* caught at Yonaguni Island. **a** Early perinucleolus stage, **b** late perinucleolus

stage, **c** yolk vesicle stage, **d** primary yolk stage, **e** secondary yolk stage, **f** tertiary yolk stage, **g** hydrated stage, **h** atretic stage, **i** postovulatory follicles (arrows). Scale bars are 500 μ m

three tertiary yolk ovaries collected in July and September. Spawning frequency and interval at Yonaguni Island were 0.12 (3/25) per day and 8.33 (25/3) days, respectively, in May–September. Atretic ovaries were observed for eight females in January, February, and April–July. Seasonal changes in the frequencies of the histological testicular classes were not distinct, and most testes had a large amount of spermatozoa throughout the year. Class 3 testes were dominant in January–September, but class 4 testes were dominant in October–December.

Mean GSI (years pooled) of females was relatively high (>0.80) in May–August and peaked in May and July (Fig. 7). Females with high GSI (>1.80) were frequently

observed in May–September. Mean GSI of males was high (>0.70) in May–September and low (<0.50) in other months.

Monthly changes in body condition

The mean ($\pm$ SD) LJFL of all female blue marlin sampled in May–September (years pooled) was 238 ± 24 cm, which was slightly but significantly larger (t test: $t = 2.22$, $df = 984$, $P < 0.05$) than the mean LJFL (235 ± 22 cm) for the other months pooled. Mean LJFL of males in May–September (years pooled) was 188 ± 11 cm, which was significantly smaller ($t = 4.98$,

Fig. 5 Photomicrographs of four testicular maturity classes of blue marlin *Makaira nigricans* caught at Yonaguni Island. **a** Class 1, **b** class 2, **c** class 3, **d** class 4. Scale bars are 500 μ m

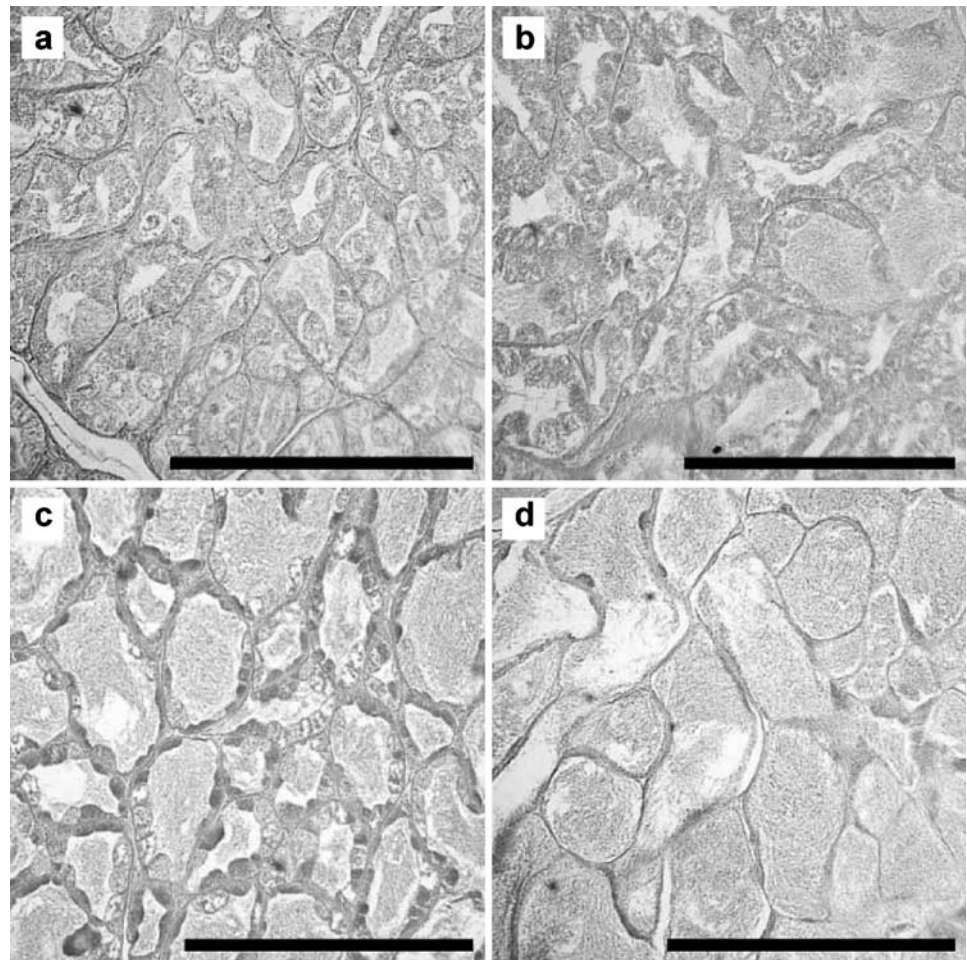


Table 1 Mean gonadosomatic indices (GSI) and mean condition factors for each gonadal class of blue marlin *Makaira nigricans* collected at Yonaguni Island between February 2003 and February 2005

Histological class	<i>n</i> (%)	Mean GSI (95% confidence interval)	Mean ($\pm$ standard deviation) condition factor
Female			
Perinucleolus	538 (75.2)	0.58 (0.56–0.60)	4.78 $\pm$ 0.46
Yolk vesicle	116 (16.2)	0.72 (0.67–0.77)	4.81 $\pm$ 0.47
Early yolk	19 (4.0)	1.36 (1.20–1.52)	4.65 $\pm$ 0.42
Tertiary yolk	23 (3.2)	2.37 (1.86–2.88)	4.61 $\pm$ 0.35
Hydrated	3 (0.4)	2.92 (1.14–4.69)	3.76 $\pm$ 0.65
Atretic	8 (1.0)	1.00 (0.72–1.28)	4.70 $\pm$ 0.16
Male			
Class 1	5 (1.3)	0.48 (0.19–0.77)	4.64 $\pm$ 0.03
Class 2	45 (11.7)	0.48 (0.37–0.60)	4.74 $\pm$ 0.42
Class 3	215 (56.0)	0.53 (0.49–0.57)	4.66 $\pm$ 0.41
Class 4	119 (30.1)	0.48 (0.43–0.53)	4.67 $\pm$ 0.44

$df = 491$, $P < 0.001$) than the mean LJFL (193 ± 12 cm) for the other seven months.

The mean condition factors differed significantly among the various ovarian maturity classifications (one-way analysis of variance: $F = 4.24$, $df = 5$, $P < 0.001$), being lower for females with yolked and hydrated ovaries than

females with un yolked ovaries (Table 1). In contrast, the mean condition factor did not differ significantly among the different male testicular maturity classes ($F = 0.52$, $df = 3$, $P = 0.67$). The monthly mean condition factors (years pooled) for females and males were significantly different (female, $F = 3.12$, $df = 11$, $P < 0.001$; male,

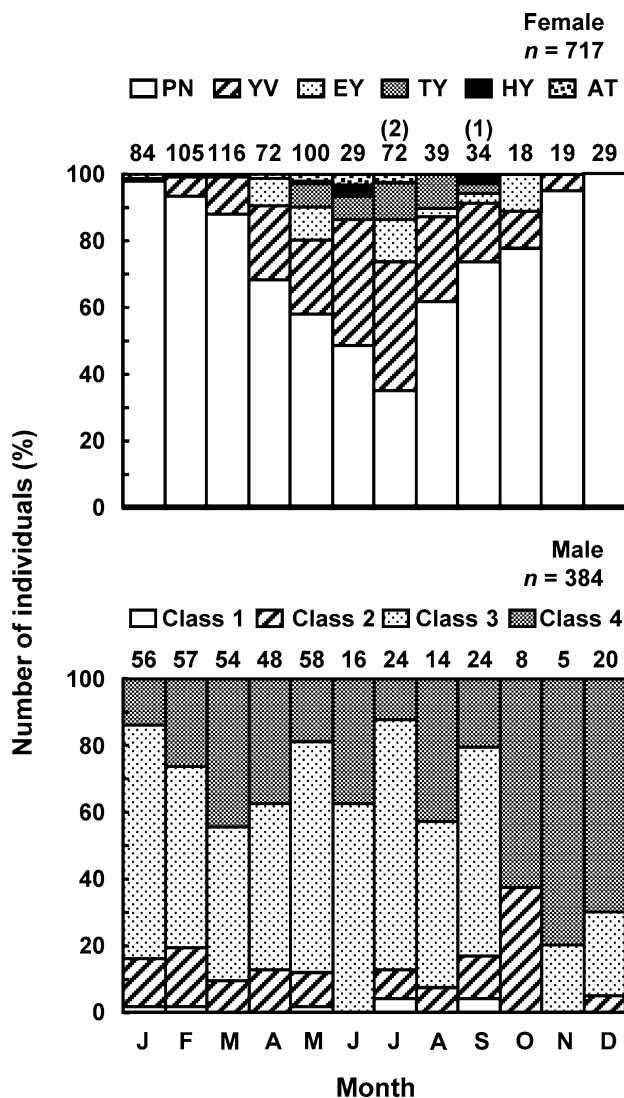


Fig. 6 Monthly changes in the frequencies of the gonadal maturity classes of blue marlin *Makaira nigricans* collected at Yonaguni Island between February 2003 and February 2005 (years pooled). Numbers on bars indicate sample sizes. Numbers in parentheses indicate numbers of ovaries with postovulatory follicles. PN, perinucleolus; YV, yolk vesicle; EY, early yolk; TY, tertiary yolk; HY, hydrated; AT, atretic

$F = 7.35$, $df = 11$, $P < 0.001$), but those of both sexes tended to decrease from March to June or July and increase from August to March (Fig. 8).

Maturation size

The smallest female with mature ovaries was 204 cm LJFL (180 cm EFL, 86 kg PW). A smaller female, 183 cm LJFL (159 cm EFL, 50 kg PW), had an early yolk ovary. All males had spermatozoa in their testes; the smallest male was 160 cm LJFL (140 cm EFL, 34 kg PW) and had class 3 testes.

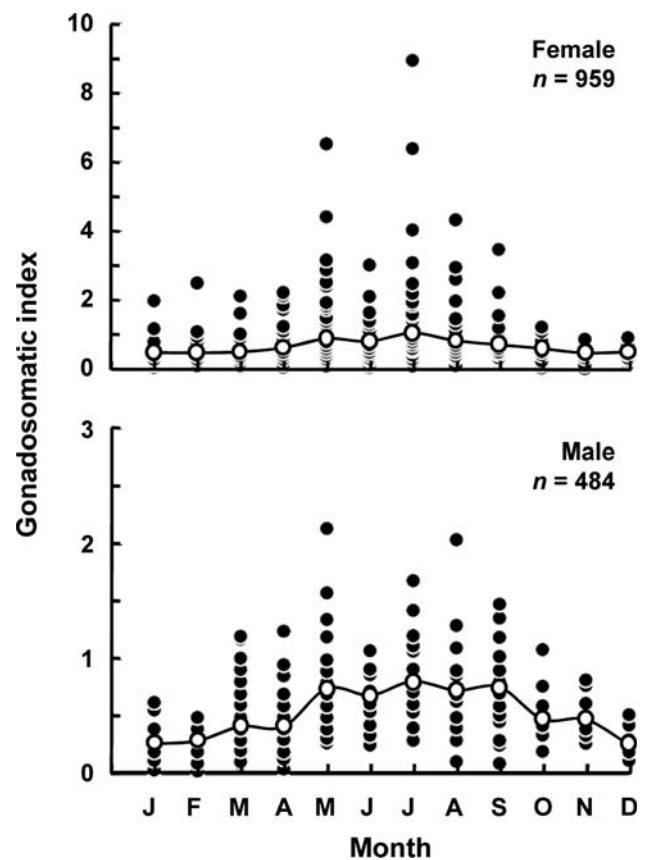


Fig. 7 Monthly changes in the gonadosomatic indices of blue marlin *Makaira nigricans* caught at Yonaguni Island between February 2003 and February 2006 (years pooled). Monthly mean values (open circles) are connected by lines

Oocyte sizes and fecundity

The size frequency distribution of the oocyte diameters of mature ovaries was trimodal, consisting of the perinucleolus, tertiary yolk, and hydrated oocyte modes (Fig. 9a). The mean ($\pm$ SD) diameter of the middle modal group was $634 \pm 65 \mu\text{m}$, and that of the largest modal group was $996 \pm 258 \mu\text{m}$. The difference between the diameters of these oocytes and those of oocytes in the histological sections may have been due to shrinkage during the processing of tissues for slide preparation and the fact that oocytes were not always sliced through their centers. The mean diameter of free eggs in the ovarian lumen was $1,187 \pm 43 \mu\text{m}$ (Fig. 9b).

The fecundities of 11 females (204–246 cm LJFL, 72–120 kg PW) ranged from 1.89 to 16.54 million eggs (Table 2). The batch fecundities of three females of 225 cm (72 kg), 232 cm (75 kg), and 238 cm (112 kg) were 0.98, 1.02, and 5.85 million eggs per fish, respectively.

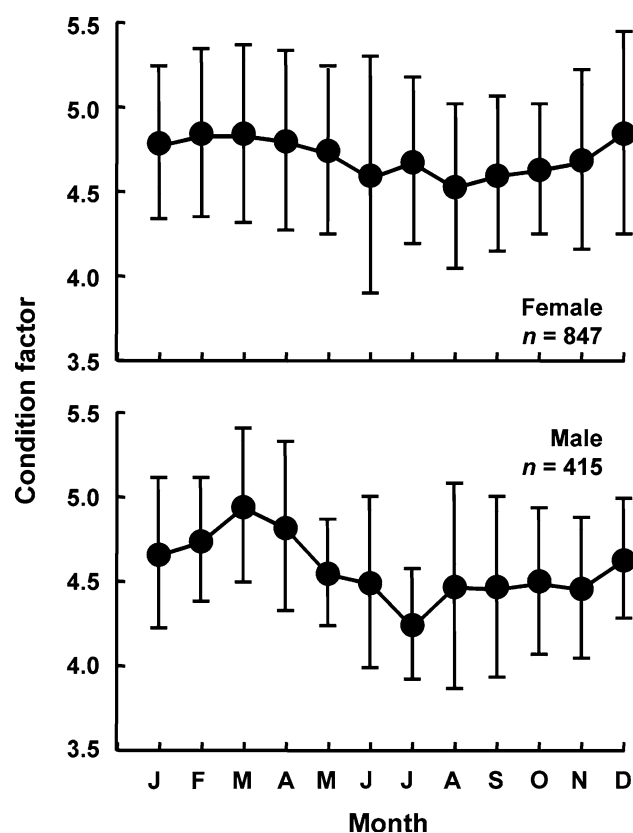


Fig. 8 Monthly changes in the mean ($\pm$ SD) condition factors of blue marlin *Makaira nigricans* collected at Yonaguni Island between February 2003 and February 2006 (years pooled). Vertical bars indicate standard deviations

Discussion

Spawning seasonality

During the study period, 25 blue marlin females with mature ovaries were caught between May and September, a period that appears to be their spawning season near Yonaguni Island. A few of these mature ovaries contained hydrated oocytes, and POF were found in other mature ovaries. Most males had testes with a large amount of spermatozoa throughout the year. Therefore, these data strongly suggest that the waters of Yonaguni Island are a spawning ground for blue marlin in the North Pacific Ocean. The spawning season for blue marlin was reported to extend from May to July east of Luzon [17] and from April to July around Taiwan [18]. Therefore, larval blue marlin near Taiwan in October may be transported from a southern spawning ground by the Kuroshio Current [20]. Our results, combined with current knowledge of larval blue marlin growth rates [29], suggest that blue marlin may spawn in September at Yonaguni Island and could be the source of larvae in Taiwanese waters in October. The theoretical spawning ground south of Taiwan, which was

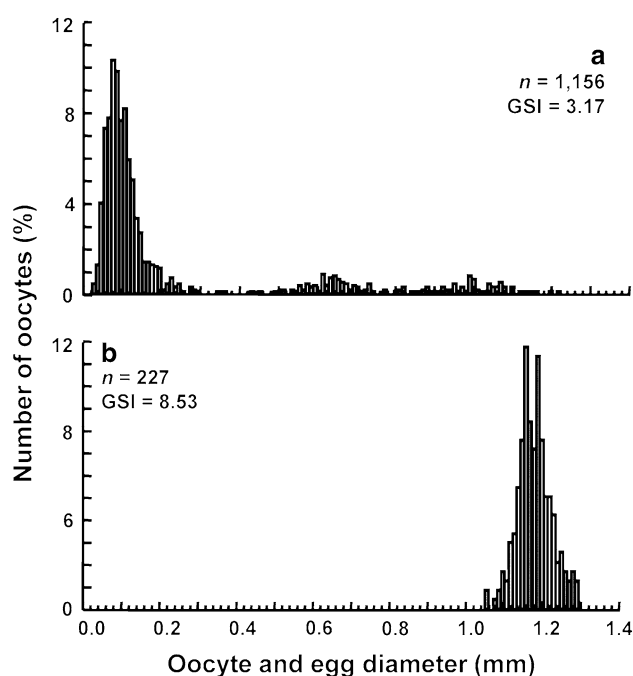


Fig. 9 Size frequency distribution plots of **a** all oocyte diameters in a subsample taken from a mature female blue marlin *Makaira nigricans* and **b** free egg diameters in a subsample of another mature female blue marlin landed at Yonaguni Island. GSI gonadosomatic index

proposed to provide the blue marlin larvae transported by the Kuroshio Current to Taiwanese waters, has not been found.

The spawning frequency (every 8.33 days) of blue marlin from May to September at Yonaguni Island appears low compared to sailfin *Istiophorus platypterus* (1.89 days) [30] and swordfish (3.02 days) [31]. The long interval between spawning may be due to the small number of ovaries with POF. The low spawning levels reported herein could reflect a bias in fishing methods that result in a low catch of spawners [15, 32], or perhaps Yonaguni Island represents the northwestern fringe of the blue marlin spawning ground in the western North Pacific [21, 33] and spawning rates may be lower at the fringe.

The GSI is a less accurate but useful indicator of spawning activity than histological examination [34]. Early reproductive studies of billfish involved GSI calculations using ovary weight and fish length, because obtaining fish weights at sea was difficult [11, 35]. In our study, individual females and males with relatively high GSI occurred in the spawning season at Yonaguni Island in May–September. Most females with a GSI of >1.80 were mature because their ovaries contained either tertiary yolk or hydrated oocytes that had completed yolk accumulation. Therefore, the presence of females with GSIs >1.80 would suggest that spawning was occurring. Oocytes >0.60 mm in diameter (tertiary yolk stage) had completed yolk accumulation, so ovaries containing oocytes >0.60 mm in

Table 2 Fecundities and batch fecundities (millions) of 11 female blue marlin *Makaira nigricans* collected at Yonaguni Island

Month specimen collected	LJFL (cm)	PW (kg)	GW (kg)	GSI	Fecundity	Batch fecundity
March	244	120	1.95	1.62	4.58	
May	204	86	2.72	3.16	6.14	
May	232	75	2.38	3.17	3.83	1.02
May	246	106	3.05	2.88	8.95	
June	225	72	1.52	2.11	2.7	0.98
July	219	80	1.83	2.29	3.33	
July	225	94	6	6.38	14.12	
July	230	100	0.97	0.97	1.89	
July	233	114	4.6	4.04	10.5	
July	238	112	9.55	8.53	16.54	5.85
August	240	108	2.82	2.61	8.07	
Mean	231	97	3.4	3.43	7.33	2.62

LJFL, lower jaw-fork length; PW, processed weight; GW, gonad weight; GSI, gonadosomatic index

diameter could also be classified as mature. Testes maturity classification, determined by comparing the relative spermatozoa area of cross-sections of testes, has successfully been used for coastal fish species [27, 28]. However, in our study, males had a large amount of spermatozoa in their testes year round in every study year, similar to Atlantic billfishes [15], and therefore were capable of spawning throughout the year. After the spawning season, spermatogenesis continued despite a decrease in spermatogonia and an increase in class 4 testes in October–December. These findings imply that the spawning area and season of blue marlin may be larger than what our data suggest. Mean monthly GSI of male blue marlin was >0.7 in May–September, the same months in which females showed high GSIs. Therefore, the mean monthly GSI for male blue marlin could be used to define the spawning magnitude or season when it can be calculated and compared.

The monthly mean condition factor of female blue marlin tended to decrease from March to June, apparently corresponding to the spawning season. The decrease from March to April might have resulted in the development of unyolked oocytes and early yolked oocytes, while that observed in May through June may have been due to continued oocyte production and spawned eggs [36]. The condition factor began to increase in September, even though the fish continued their reproductive activities, because their bodies may have begun to reclaim energy through the resorption of oocytes that were not going to be spawned. The monthly mean condition factor of males showed a steeper decline than that of females during the same period. The energy requirements for gamete production are thought to be less for males than females, so males must expend more energy on courtship and posturing

to establish dominance. From March to June or July, there was a decreasing trend in condition factor while spawning. Black marlin *Istiompax indica*, which spawn mainly in the southern hemisphere and not at Yonaguni Island [37, 38], show an increasing trend in condition factor during those months while on their seasonal visit to Yonaguni Island [39]. Condition factor may be affected by feeding intensity [22]. Because the feeding activities of both marlin species at Yonaguni Island are similar [19, 39], the difference in condition factor trends between the two species might be due to differences in their spawning activities.

Reproductive characteristics

Females outnumbered males at Yonaguni Island throughout the study period. The sex ratio is about equal during the spawning season at Puerto Rico and the Virgin Islands [14], but males dominate during the spawning season in Hawaiian waters [16]. The dominance of males in Hawaii suggests that multiple males chase and court a single female [16]. Although females outnumbered males at Yonaguni Island during the spawning period, there was a relatively low abundance of mature females and a higher proportion of sexually active and mature males in the population. A similar stock structure of reproductive blue marlin was observed off the island of Hawaii [16]. Differences in sex ratio have been attributed to differential migration and distribution patterns between sexes and age groups of blue marlin and swordfish [9, 11, 40]. Sex ratios related to reproductive behavior should be analyzed using only sexually mature fish.

The smallest female with an ovary containing tertiary yolk stage oocytes was 204 cm LJFL. The smallest female with an ovary containing early yolk stage oocytes was 183 cm LJFL. Oocytes in the early yolk stage are thought to develop rather rapidly with no noticeable growth in body length, so the minimum length at maturity was considered to be 183 cm LJFL (159 cm EFL). Various authors have reported the smallest size of mature female blue marlin for their region by various criteria [11, 12, 14–16]. Except for the eastern Pacific [11], the minimum mature lengths are larger in other regions compared to that for Yonaguni Island. The smallest mature female in the eastern Pacific was 155 cm EFL [11], comparable to our results.

The smallest male, 160 cm LJFL, already had a large amount of spermatozoa in the testes, so the length at first maturation is likely to be less than 160 cm LJFL (140 cm EFL). The precise size could not be determined due to the lack of small males in our samples, resulting either from the size selectivity of the bait-trolling fishing method or the absence of small males at Yonaguni Island [9].

Hydrated oocytes (996 μm) and free eggs in the ovary (1,187 μm) were slightly smaller than fertilized eggs (1.27 mm) [41], and were probably just about to be

spawned. The batch fecundity ranged from 0.98 to 5.85 million eggs for females 225–238 cm LJFL (72–112 kg PW), similar to that of swordfish off Florida (1.40–4.22 million eggs for 69–268 kg BW) [42] and eastern Australia (1.17–2.50 million eggs for 99–243 kg BW) [31].

Fecundity ranged from 1.89 to 16.54 million eggs for female blue marlin 204–246 cm LJFL (72–120 kg PW), i.e., less than the 31.5–98.9 million eggs estimated for blue marlin 127–420 kg BW in Hawaiian waters [16]. The difference in fecundity estimates may be due to a difference in criteria for selecting developmental oocyte stages to be counted. Annual fecundity is properly estimated by batch fecundity, spawning frequency, and spawning duration [43]. Annual fecundity could not be determined in our study because of the restricted sampling area for such a highly migrational species. Our fecundity estimates represent the minimum potential number of eggs produced in a spawning season.

Migration patterns

The number of blue marlin landed usually increases from March through July in most years at Yonaguni Island [9], but it increased starting in January or February in 2003 and 2004. Their early arrival appeared to be related to water temperature or unknown oceanographic conditions. The number landed per month decreased after spawning in August or October, and blue marlin eventually appeared to leave the area. Blue marlin are thought to migrate to the southern hemisphere during the austral summer, October to March [7, 8], and their body condition recovers prior to the next spawning season in the western North Pacific. Recovery of their condition during the non-spawning season in the northern hemisphere implies that they do not spawn twice in a year, i.e., once in the northern hemisphere and again in the southern hemisphere.

Size differences in males were observed between the spawning and non-spawning seasons, which may be due to the influx of small male recruits in the spawning season [9]. The distribution patterns of different size classes observed for blue marlin in the eastern Pacific and swordfish in the north and south Pacific may be related to spawning activity [11, 31, 40]. Recruitment of small but mature male blue marlin in the spawning season may reflect their polyandric mating system.

We determined that blue marlin migrate near Yonaguni Island for feeding throughout the year [19], and part of the population was observed to spawn at this location between May and September. However, further research on the spawning seasonality and frequency in other regions is necessary for a comprehensive understanding of blue marlin reproductive biology and related migration patterns in the Pacific Ocean.

Acknowledgments We are grateful to K. Agena, H. Agena, and the fishermen of Yonaguni Island for recording data and helping with the sampling of blue marlin. We thank J. Uchiyama, R. Humphreys Jr. (NOAA, National Marine Fisheries Service) and anonymous reviewers for reviewing and improving the manuscript. This research was supported by a grant from the Japan Fisheries Agency, and partially supported by the 21st Century COE program of the University of the Ryukyus.

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